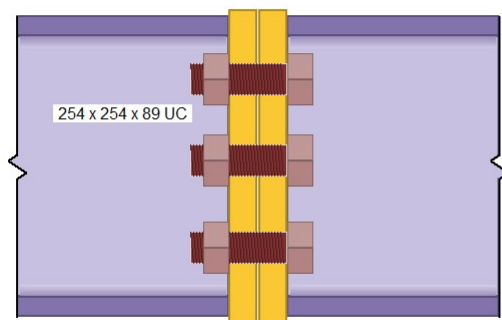
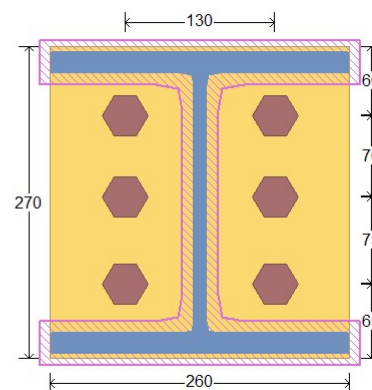


End plate splice calculation

Beam: Lower Ground Spine Beam

Beam span: 6.6 m. Section: **254 x 254 x 89 UC** S355

End plates: 260 x 270 x 25, M27 8.8 bolts (6)



Splice 1: 2.2 m. from R1. BM: 126.3 kNm SF: 29.10 kN (factored)

Section size: 254 x 254 x 89 UC S355

Section dims: D = 260.3 B = 256.3 T = 17.3 t = 10.3 d = 200.3 r = 12.7

Use S275 end plates 260 x 270 x 25 mm

Use 3 pairs of M27 8.8 bolts at 130 mm cross centres; top of plate to top row of bolts: 60; second row: 130 mm; rows 3- spaced at 75 mm

Min fillet weld sizes: Top (compression) flange: nominal 8 mm (bearing fit assumed)

Bottom (tension) flange: $17.3/(2 \times 0.7) = 12.4$ mm, say 13 mm (full strength)Web: $10.3/(2 \times 0.7) = 7.4$ mm, say 8 mm (full strength)

N.B. 13mm welds may not be practicable

Basic detailing checks

Using M27 bolts, washer diameter 50 mm

Check plate height (270 mm) $\geq$ section depth (260 mm) OKCheck top bolt washer clears top flange weld: top of plate to top bolt (60 mm) $\geq$ 56 mm OKCheck bottom bolt washer clears bottom flange weld: top of plate to bottom bolt (205 mm) $\leq$ 209 mm OKCheck bolt washers clear web welds: bolt cross centres (130 mm) $\geq$ 77 mm OKCheck bolt cross centres $\geq$ 2.5D (67.5 mm) OKCheck bolt vertical centres (70 mm) $\geq$ 2.5D (67.5 mm) OKBolt edge distance (65.0 mm) $\geq$ 1.25D (33.8 mm) OKTop bolt end distance (60 mm) $\geq$ 1.25D (33.8 mm) OKBottom bolt end distance (65 mm) $\geq$ 1.25D (33.8 mm) OK

All dimensional checks satisfied

End plate beam splice**Section size: 254 x 254 x 89 UC****Basis of design**

This calculation is derived from the SCI publication : Design of Moment Connections. The compression zone is taken as acting at the centre of the top flange. The top pair of bolts act in shear only. The bending moment at the splice is resisted by the other bolt rows. The tensile capacity of each bolt pair is the least value of:

* The capacity of the bolt row acting alone [Table 2.5]

* Other rows: the capacity of this bolt row and the row(s) below it acting as a group
less the capacity of the row(s) below it acting as a group (2.3.1)

* The tensile capacity of the beam web (row(s) above bottom row)

Bolt bearing capacity on plate = $27 \times 25 \times 460/1000 = 310.5$ kN

Bolt shear capacity = $459 \times 375/1000 = 172$ kN <<<

Enhanced bolt tension capacity, $P_t' = 459 \times 560/1000 = 257.0$ kN

Max tensile length of web per bolt row, $L_t = (130/2) \times \tan 60 = 113$ mm above and below each row

N.B. the length below may not extend into the length above from the row below

Web tensile capacity per mm = $10.3 \times 345/1000 = 3.55$ kN/mm

m_1 , bolt centre to 20% into web weld = 53.5 mm (8mm fillet welds assumed)

m_2 , bolt centre to 20% into bottom flange weld = 32.4 mm (13mm fillet welds assumed)

$l_1 = m_1/(m_1 + e) = 0.45$ $l_2 = m_2/(m_1 + e) = 0.27$ $a = 2p$ (P207 Appendix iii)

n , effective edge distance = min {edge distance, $1.25m$ } = 65.0 mm

$L_{eff}(i) = 2pm = 335.8$ mm $L_{eff}(ii) = 4m + 1.25e = 295.0$ mm $L_{eff}(iii) = am = 335.8$ mm (Table 2.4, $a = 2p$)

L_{eff} for bolts acting alone (Table 2.5)

Bottom row: $L_{eff} = \min\{\max\{[ii+iii]/2, ii\}, i\} = 315.4$ mm

Other rows: $L_{eff} = \min\{ii, i\} = 295.0$ mm

L_{eff} for bolts acting as a group (Table 2.6)

Bottom row: $L_{eff} = \max\{ii/2, iii/2\} + p/2 = 205.4$ mm

Non-bottom end row: $L_{eff} = ii/2 + p/2 = 185.0$ mm

$M_p = L_{eff} \times t^2 \times p_y / 4$

Row capacity

Mode 1 (end plate yields at bolt positions and either side of web): $P_r = 4M_p/m$

Mode 2 (end plate yields either side of web; bolts fail in tension): $P_r = (2M_p + n(SP_t'))/(m + n)$

Mode 3 (bolt tensile failure): $P_r = SP_t'$

Bolt rows are numbered from the bottom (bottom pair = 1; top pair = 3)

Row	L_{eff}	M_p	Mode 1	Mode 2	Mode 3	$L_{t,web}$	$P_{t,web}$	$P_{t,group}$ kN	P_t kN
1	315	13,061	977	503	514				502.6<<<
2	295	12,217	914	488	514	225	800		488.4
[1+2]	390	16,167	1,210	837	1,028			837	
[1+2]-1									334.5<<<

Check moment and shear capacity of splice

Compressive resistance of top flange = $1.4 \times 345 \times 256.3 \times 17.3/1000 = 2142$ kN

Bolt line	Bolt LA mm	P_r kN	Row RM kNm	Bolt F_v/P_v	F_v kN
1	192	502.6	96.3	0.40	137.7
2	116	*305.8	35.6	0.40	137.7
3	46	0.0	0.0	1.00	344.3
Totals		808.4	131.9		619.7

* tension reduced to be within elastic limit

Section ultimate moment capacity = $p_y S_x = 421$ kNm; Splice capacity = 131.9 kNm (31.3%)

Splice 1: Design moment: 126.3 kNm Design SF: 29.1 kN

No further checking necessary